Catalytic Hydrogenation of Carbon Dioxide Using Heterogeneous Ruthenium Complexes

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ABSTRACT

Direct conversion of carbon dioxide to methanol requires high-selectivity heterogeneous catalysts.

METHODOLOGY

We synthesized ruthenium nanoparticles anchored on silica supports.

LIMITATIONS

Catalyst deactivation occurs after 48 hours of continuous operation due to carbon fouling.

FUTURE WORK

Regeneration protocols using supercritical solvents will be tested.

REFERENCES

1. Richter et al. 2024.